



Translating sign language into speech

Not everybody can understand sign language – this is a practical difficulty that hearing and speech impaired people face everyday. Four students from Ukraine have developed a simple and inexpensive solution called Enable Talk comprising special gloves and a mobile application, which can detect signs made by the user wearing the gloves, and convert these into text or speech on the phone. Working on the user feedback, they are now enhancing the system to also track arm movement in addition to just fingered signs.

Inside: Eleven flex sensors and eight touch sensors embedded in the glove convert movement into proportionate electrical resistance. The glove also includes 3D digital linear acceleration sensor, magnetic sensor, accelerometer and gyroscope. An Atmel XMEGA A3 microcontroller processes analogue signals from sensors and transmits the resulting data to a mobile using Class 2 Bluetooth module (BlueCore4-Extreme chipset from Cambridge Silicon Radio). The glove is powered by a 750mAh lithium-polymer accumulator, which can be charged through USB or built-in solar battery.

Easy and intelligent LED bulbs

You can control these light-emitting diode (LED) bulbs with your mobile phone – and no, we are not talking of complex building-wide lighting solutions. iLumi is a simple LED-based bulb that can be screwed into any common socket. It is available in small, large, white and full-spectrum models, which are supposedly super energy-efficient, last up to 20 years, and can be controlled with a simple mobile application. You can schedule the light, alter its brightness of hue, motion-activate it, and do much more. Founders Corey Egan and Swapnil Bora are waiting to finalise the design and manufacture the bulbs, which have already won numerous awards.

Inside: iLumi's HyperLux technology allows it to be programmed to display millions of colours and makes it one of the brightest bulbs around. The bulb, which is compatible with normal sockets and power supply, includes Bluetooth, 16-bit, 256kB flash processor, additional 4-megabit flash memory, real-time lighting control for up to 100 iLumi bulbs, and clock with battery back-up. The control application is simple and intuitive, and can be downloaded and run on Android and iOS devices. The company plans to include a software developer kit (SDK), to come up with innovative ways for programming and controlling the bulb.

